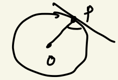
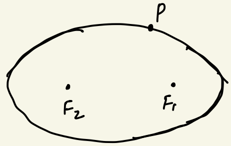
□ Straight-edge + Compass + Pencil

↓

(Unmarked ruler)

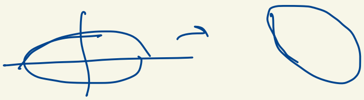


Arbitrarily Oriented Ellipse is given

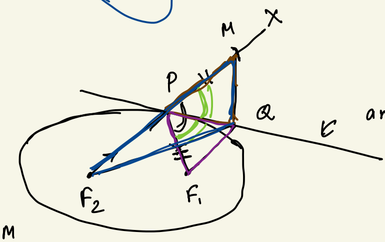


P is any pt. on the ellipse.

F_1, F_2 : Foci



Draw a tangent to the ellipse at P



angle bisector of $\angle F_1PX$

tangent to the ellipse at P.

$PM = PF_1$

Join MQ

$\triangle PQF_1$ and $\triangle PQM$

$PF_1 = PM$ (cnt)

$\angle QPF_1 = \angle QPM$

PQ Common

↓

$\triangle PQF_1 \cong \triangle PQM$ (S-A-S)

$QF_1 = QM$

Take any other point Q ($\neq$ P) on the st. line (external angle bisector)

We shall show that Q does not lie on the ellipse

$\triangle$-ineq in $\triangle MQF_2$

$QF_2 + MQ > MF_2$

$\Rightarrow QF_2 + QF_1 > \boxed{PF_2 + PF_1 = \text{const}}$

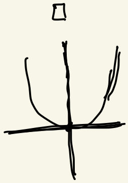
$\Rightarrow$ Q cannot lie on the ellipse.

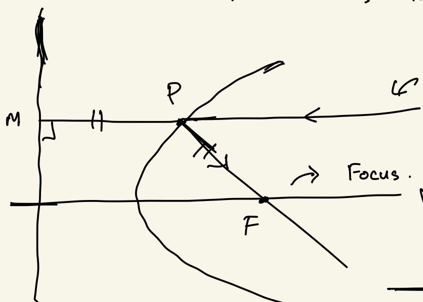
Since the choice of Q ($\neq$ P) was arbitrary,

$\Rightarrow$ The only pt the Ellipse and the st line have in common is P

$\Rightarrow$ Line is tangent to the ellipse at P. □

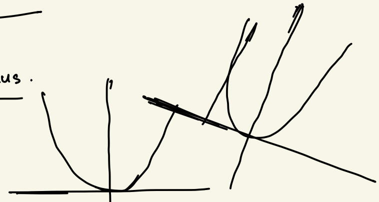
□



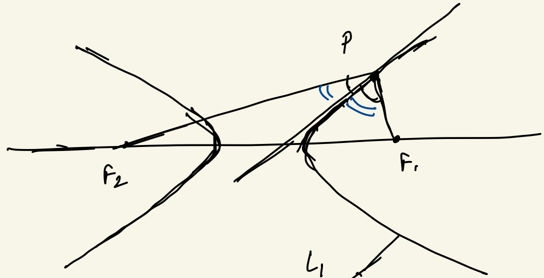


Line || to the axis of symmetry

Focus.

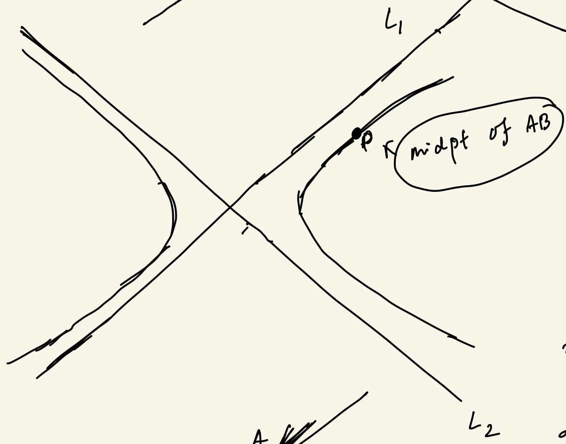


☐



$|PF_2 - PF_1| = \text{const}$

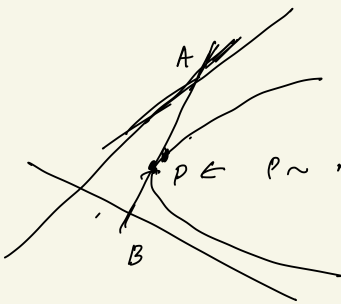
☐



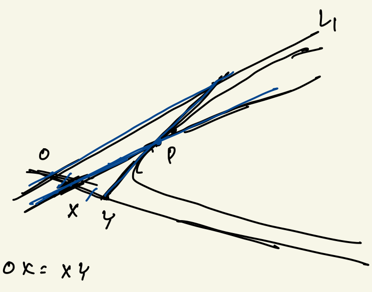
A hyperbola and its two asymptotes are given

P is an arbitrary pt on the hyperbola.

Draw the tangent at P to the hyperbola.



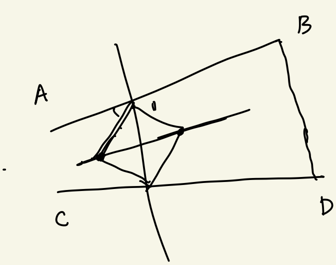
← P ~ midpt of AB



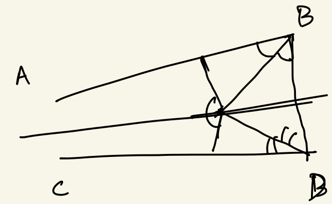
midpt Theorem

OX = XY

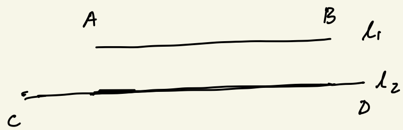
☐



Without extending draw the angle bisector b/w the line BA, CD



Only Straight-edge and a pencil (No compass).

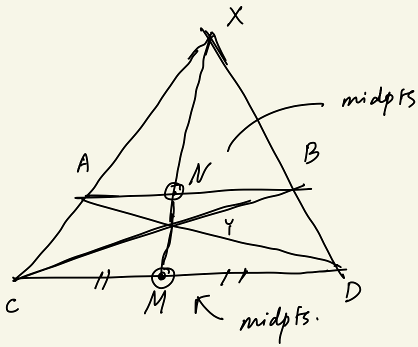


$l_1 \parallel l_2$ given. Using st. edge, pencil only, find the mid pts of l_1, l_2

$AN \parallel CM$

$\Rightarrow \triangle XAN \sim \triangle XCM$

$\Rightarrow \frac{AN}{CM} = \frac{XN}{XM}$



$\triangle XBN \sim \triangle XMD$

$\Rightarrow \frac{BN}{MD} = \frac{XN}{XM}$

$\frac{Ax}{AC} = \frac{XB}{BD}$

$\Rightarrow \frac{Ax}{AC} \cdot \frac{BD}{XB} = 1$

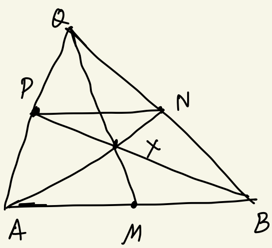
Ceva ↘

$$\frac{Ax}{AC} \cdot \frac{CM}{MD} \cdot \frac{BD}{XB} = 1 \Rightarrow \boxed{CM = MD}$$

$\frac{AN}{CM} = \frac{BN}{DM}$

$\Rightarrow \boxed{AN = BN}$

□



Given: St line AB

Midpt of AB is also given: M.

P ~ external to AB

Using only st-edge draw a line via P $\parallel$ to AB.

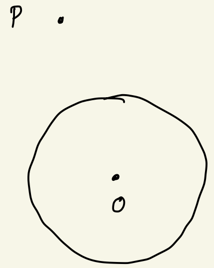
Join AP $\xrightarrow{\text{extend}}$ Q

Join BP, Join MQ $\Big\}$ $BP \cap MQ = X$

Ax ⟶ cuts BQ at N

$PN \parallel AB$.

HW



a circle with centre O - given.

P is any pt outside the circle.

Using st-edge + Pencil only (No compass ~~are~~ allowed) draw tangents from P to the circle.

$\boxed{\text{Steiner:}}$ Any geometric construction that can be done with compass + st. edge, can also be done using a st. edge only, if a circle with centre is already provided.